

Project Documentation

1. Project Planning

The main objective of the project was to build a **dashboard analyzing the unemployment rate in the USA**, identifying its causes, and understanding the most in-demand jobs.

The project was divided into multiple stages to ensure structured execution:

1. **Data Collection:** Searching for and obtaining real datasets relevant to unemployment.
2. **Data Cleaning:** Removing missing or inconsistent entries and preparing the dataset for analysis.
3. **Chart Planning & Visualization Design:** Conceptualizing the types of charts to use and determining the insights they should provide.
4. **DAX Measures & Calculations:** Creating calculated measures in Power BI to compute key metrics such as employed, unemployed, outside labor force, and unemployment rate.
5. **Data Presentation & Dashboard Design:** Designing the dashboard layout, arranging charts and cards for maximum clarity and usability.

A **timeline** was created to allocate tasks efficiently across the team. Each stage was assigned a specific duration (e.g., 3 days for data collection and 3 days for data cleaning), while the more complex stages such as DAX measures and dashboard design were given more time as needed.

Progress was tracked regularly by checking in with team members responsible for each task and ensuring completion according to the deadlines set. This structured planning ensured timely delivery and smooth coordination among the team.

2. Stakeholder Analysis

The stakeholders of the project were identified to ensure the dashboard met the needs of all interested parties. The main stakeholders included:

1. **Evaluating Committee:** Required a clear, easy-to-understand overview of unemployment trends, along with detailed analyses when needed. They focused on accuracy, readability, and actionable insights.
2. **End Users:** Needed quick access to key metrics, the ability to filter data by year, month, and state, and visualizations that made it easy to compare employment, unemployment, and labor force statistics.
3. **Project Team:** Needed clear task assignments, timelines, and access to prepared data and DAX measures to complete their responsibilities efficiently.

The analysis involved understanding each stakeholder's needs and designing the dashboard accordingly:

- **Overview Page** for quick insights.
- **Detail Page** for in-depth analysis.
- **Interactive Filters and Navigation Buttons** to allow stakeholders to explore the data as required.

Regular communication with the team ensured all tasks were completed on time and that the dashboard met the expectations of both the committee and the end users. This structured approach to stakeholder analysis helped guarantee the project's success and relevance to all parties involved.

3. Database Design

The project database was designed in an organized manner to support efficient and flexible analysis of unemployment data.

- The data is structured in **interconnected tables** to facilitate analysis in Power BI, with the ability to add new metrics or dimensions in the future (e.g., age groups, industries, or education levels).
- **Key DAX measures** were created to calculate total employed, total unemployed, outside labor force, unemployment rate, and any additional measures as needed.
- Relationships between tables were designed to ensure smooth operation of filters and slicers, allowing users to navigate pages and data easily.
- The data was cleaned and validated to ensure **data integrity and accuracy**, including the removal of missing or inconsistent values.

This design is flexible, allowing for the integration of additional tables or metrics in the future without restructuring the existing database.

4. UI/UX Design

Current Dashboard Design

- **Overview Page:** Displays key indicators such as employed, unemployed, outside labor force, and unemployment rate using Cards. Includes a Map Chart for unemployment rate by state, a Line Chart showing trends over time, and Bar Charts highlighting top and bottom 3 states.
- **Detail Page:** Provides deeper insights with charts broken down by year, month, state, and labor status. Includes additional Pie/Donut Charts, Line Charts showing relationships

between population growth and unemployment rate, and Bar Charts for comparison across statuses.

User Experience (UX) Features

- Clear **colors and labels** for easy interpretation.
- Logical arrangement of cards and charts, placing the most important metrics at the top.
- Slicers allow users to filter data dynamically by year, month, state, and labor status.
- Navigation buttons facilitate smooth page transitions.

Future Flexibility

- Additional dashboards can be designed to show:
 - Jobs per country, average salaries, and the number of people employed in each role.
 - Reasons for job rejections or terminations, with trend analysis by year or state.
- All charts, cards, and slicers are **designed to be dynamic and easily adjustable**, enabling seamless integration of new data, metrics, or visualizations without redesigning the entire dashboard.

Note: Every element in the dashboard, whether a chart, card, or filter, is **modifiable and updatable** as the project evolves.
